

CLINICAL STUDY PROTOCOL

1. Study Title

A Phase IIb, Randomized, Double-Blind, Placebo-Controlled Study Evaluating the Efficacy and Safety of IMT-201 in Patients with Type 2 Diabetes Mellitus

Protocol Number: MOCK-METABOLIC-001

Version: 1.0

Date: 2026-01-01

2. Study Objectives

Primary Objective:

- To evaluate the efficacy of IMT-201 on HbA1c reduction in patients with type 2 diabetes mellitus

Secondary Objectives:

- To evaluate the safety and tolerability of IMT-201
- To evaluate the effect on fasting glucose, lipid panel, and body weight

3. Eligibility Criteria

3.1 Inclusion Criteria

- Male or female age 18-75 years old
- Confirmed diagnosis of Type 2 Diabetes Mellitus for at least 12 months
- HbA1c between 7.0% and 10.0% at screening
- Body Mass Index (BMI) between 25 and 45 kg/m²
- Stable metformin therapy (≥1000 mg/day) for at least 8 weeks prior to screening
- Willing to provide written informed consent

3.2 Exclusion Criteria

1. Type 1 Diabetes or history of diabetic ketoacidosis
2. Any history of acute or chronic pancreatitis
3. Severe hepatic impairment (ALT/AST >3x upper limit of normal)
4. eGFR <30 mL/min/1.73m²
5. Current use of GLP-1 receptor agonists, SGLT2 inhibitors, or insulin therapy (excluding basal insulin for rescue)
6. Pregnant or nursing women

4. Study Design

This is a 24-week (168 days) randomized, double-blind, placebo-controlled study with the following visits:

Visit	Study Day	Window	Key Procedures
Screening (V1)	Day -14 to 0	±7 days	Labs, ICF, physical exam, eligibility confirmation
Baseline (V2)	Day 0	±3 days	Randomization, vital signs, IP dispensation, PK sample (trough)
Week 4 (V3)	Day 28	±3 days	Safety labs, AE review, PK sample (trough)
Week 12 (V4)	Day 84	±5 days	HbA1c, lipid panel, CMP, fasting glucose, PK sample (trough), safety labs
Week 20 (V5)	Day 140	±5 days	Safety labs, AE review, IP accountability
End of Study (V6)	Day 168	±7 days	Final HbA1c, lipid panel, CMP, safety labs, return IP, final assessment

5. Concomitant Medication

5.1 Permitted Medications

- Metformin: stable dose (≥1000 mg/day) for ≥8 weeks prior to screening and maintained stable throughout the study

- Statins: stable dose for ≥ 4 weeks
- Antihypertensive medications: stable dose for ≥ 4 weeks
- Aspirin: ≤ 100 mg/day for cardiovascular protection

5.2 Prohibited Medications

- GLP-1 receptor agonists (e.g., semaglutide, dulaglutide)
- SGLT2 inhibitors (e.g., empagliflozin, canagliflozin)
- Sulfonylureas (e.g., glyburide, glipizide)
- Thiazolidinediones (e.g., pioglitazone, rosiglitazone)
- DPP-4 inhibitors (e.g., sitagliptin, linagliptin)
- Insulin therapy (except basal insulin for rescue therapy)
- Alpha-glucosidase inhibitors (e.g., acarbose, miglitol)
- Meglitinides (e.g., repaglinide, nateglinide)
- Any other investigational antidiabetic agents

6. Safety Monitoring and Adverse Events

6.1 Serious Adverse Event (SAE) Reporting

1. **Initial Notification:** Site must notify the Sponsor within 24 hours of awareness of an SAE
2. **Written SAE Report:** Completed SAE Form submitted to Sponsor within 5 calendar days of initial notification
3. **Regulatory Reporting:** Life-threatening or fatal SAEs reported to regulatory authority within 7 calendar days
4. **Follow-up:** All SAEs followed to resolution; final outcome reported to Sponsor

6.2 Laboratory Monitoring

- Fasting blood glucose (capillary or plasma)
- HbA1c (central lab)
- Lipid panel (total cholesterol, LDL, HDL, triglycerides)
- Comprehensive metabolic panel (sodium, potassium, chloride, CO₂, BUN, creatinine, glucose, calcium)
- Liver function tests (AST, ALT, bilirubin, albumin)
- Pancreatic enzymes (amylase, lipase) at baseline and Week 12
- Urine ketones and protein at each visit

7. Dosing and Administration

Investigational Product: IMT-201

Formulation: 50 mg oral tablet

Dose: 50 mg once daily

Duration: 24 weeks (168 days)

Route: Oral, with water, at approximately the same time each day

7.1 Missed Dose Management

1. If a dose is missed and remembered within 12 hours of the scheduled time: take the missed dose as soon as remembered
2. If a dose is missed and >12 hours have passed: skip the missed dose and continue with the next scheduled dose
3. Do NOT double the dose to make up for a missed dose
4. If the subject misses 2 or more consecutive doses, contact the Investigator for assessment before resuming
5. Document all missed doses in the subject diary

8. Data Collection

All study data will be recorded in the electronic case report form (eCRF) within 5 business days of the study visit. Source documents will be reviewed by the Clinical Research Associate (monitor) at each monitoring visit to ensure accuracy and completeness. Any discrepancies between source documents and eCRF will be resolved through formal queries.

9. References

- ICH Good Clinical Practice (E6(R2))
- FDA 21 CFR Part 312 (Investigational New Drug Application)
- ANMAT Regulaciones para Ensayos Clínicos en Argentina
- Protocol Version 1.0, dated 2026-01-01